

BSDS (2025 Semester II): Statistics II

Endsem: Part A

NAME :

ROLL :

Time: 90 minutes

Total marks: 52

Maximum attainable marks: 45

All the rough works should be done in the space provided in the question paper.

Q.I Each of the following questions has one or more correct options. Identify all of them. $6 \times 4 = 24$

1. Let $X \sim F_\theta$ where $F_\theta \equiv \text{Uniform}(0, \theta)$. Indicate the correct statements below.

- A. F_θ belongs to a scale family. ☐
- B. F_θ belongs to a location family. ☐
- C. F_θ belongs to an exponential family. ☐
- D. The distribution of $1/X$ belongs to a scale family. ☐
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2. Consider that $X \sim \text{Exponential}(\lambda)$ distribution with expectation $1/\lambda$, and a constant $\alpha \in (0, 1/2)$. Let $Y = \exp\{-X\}$ and γ_α is the upper- α point of X .

- A. The upper- α point of Y is $\exp\{-\gamma_\alpha\}$. ☐
- B. The upper- α point of X is $\exp\{-\lambda\alpha\}$. ☐
- C. The upper- α point of X is $-\{\log(1 - \alpha)\}/\lambda$. ☐
- D. $E(X)$ is the upper- α point of X , where $\alpha = \exp\{-1\}$. ☐
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3. Let $X_1, \dots, X_n \stackrel{i.i.d.}{\sim} \text{Uniform}(-\theta, \theta)$ distribution. Consider three estimators of θ , $T_{1,n} = X_{(n)}$, $T_{2,n} = -X_{(1)}$ and $T_{3,n} = Y_{(n)}$, where $Y_i = |X_i|$ for $i = 1, \dots, n$.
- A. $E(T_{1,n}) \rightarrow \theta$ as $n \rightarrow \infty$. ☐
- B. $E(T_{2,n})$ over-estimates θ . ☐
- C. $E(T_{3,n}) \rightarrow \theta$ as $n \rightarrow \infty$. ☐
- D. The squared bias of $T_{3,n}$ is always smaller than that of $T_{1,n}$ or $T_{2,n}$. ☐
- [Hint: Try to express $T_{3,n}$ as a function of $T_{1,n}$ and $T_{2,n}$.]*
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4. Let $X_1, \dots, X_n \stackrel{i.i.d.}{\sim} \text{Bernoulli}(\theta)$ distribution. Consider the problem of testing $H_0 : \theta = 0.5$ against $H_1 : \theta > 0.5$, and the test function of a level- α test

$$\phi(\mathbf{x}) = \begin{cases} 1 & \text{if } \sum_{i=1}^n x_i > c_\alpha, \\ \gamma & \text{if } \sum_{i=1}^n x_i = c_\alpha, \\ 0 & \text{otherwise,} \end{cases}$$

where $\gamma \in (0, 1)$. Let $\mathbf{y} = \{y_1, \dots, y_n\}$ be a realization of $\{X_1, \dots, X_n\}$.

- A. If H_0 is rejected given $\mathbf{y}$ at 5% level of significance, then it must be rejected at 1% level of significance. ☐
- B. Let $\sum_{i=1}^n y_i = c_\alpha$, and $p(\mathbf{y})$ be the p -value given the sample $\mathbf{y}$. Then H_0 will be accepted based on the $p(\mathbf{y})$. ☐
- C. Let $\sum_{i=1}^n y_i = c_\alpha$, then the test cannot be conducted without evaluating the p -value. ☐
- D. Let $\sum_{i=1}^n y_i = c_\alpha + 1$, then $p(\mathbf{y}) > \alpha$. ☐
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5. Let X be a random variable taking values in $\{-3, \dots, 3\}$. Consider the testing problem $H_0 : X \sim f_0$ against $H_1 : X \sim f_1$, where

x	-3	-2	-1	0	1	2	3
$f_0(x)$	0.20	0.15	0.10	0.05	0.10	0.15	0.25
$f_1(x)$	0.05	0.10	0.15	0.40	0.15	0.10	0.05

Consider the most powerful level- α test ϕ^* by Neyman-Pearson's (NP) lemma, and let $\alpha = 0.05$. Then indicate which of the following statements are true.

- A. $\phi^*(X)$ is a randomized test. ☐
- B. $\phi^*(X) = \mathbb{I}(X = 0)$, where $\mathbb{I}$ is the indicator function. ☐
- C. The probability of type II error for $\phi^*(X)$ is 0.40. ☐
- D. $\phi^*(X)$ is unbiased. ☐
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6. Let $X_n, n \in \mathbb{N}$, be a sequence of independent random variables such that $X_n \sim \text{Binomial}(m, 1/n)$, and $Y_n = \sum_{j=1}^n X_j$. Then which of the following statements are correct?
- A. Y_n is distributed as $\text{Binomial}(m, 1)$. ☐
- B. $X_n \xrightarrow{p} 0$ as $n \rightarrow \infty$. ☐
- C. $X_n \xrightarrow{L^2} 0$ as $n \rightarrow \infty$. ☐
- D. $\text{var}(Y_n) \rightarrow 0$ as $n \rightarrow \infty$. ☐
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Q.II Write the answers to the following questions in one sentence.

$7 \times 4 = 28$

1. Let $X_1, \dots, X_n \stackrel{i.i.d.}{\sim} \text{Poisson}(\lambda)$ distribution. Consider the function of λ , $g(\lambda) = \exp\{-\lambda\}$. Find an unbiased estimator of $g(\lambda)$.

[Hint: Express $g(\lambda)$ as $P(A_X)$ for some appropriate event A_X .]

2. Let $X_1, \dots, X_n \stackrel{i.i.d.}{\sim} \text{Pareto}(\xi_0, \alpha)$ distribution, with pdf

$$f_X(x) = \frac{\alpha \xi_0^\alpha}{x^{\alpha+1}}, \quad x > \xi_0 > 0, \quad \alpha > 0,$$

where ξ_0 is known. Find the MLE of α .

3. Let $X_n \sim N(\mu_n, \sigma_n^2)$ where $\mu_n \rightarrow 0$ and $\sigma_n^2 \rightarrow 0$, as $n \rightarrow \infty$. Show that $X_n \xrightarrow{L^2} 0$ as $n \rightarrow \infty$.

4. Let $X_1, \dots, X_n \stackrel{i.i.d.}{\sim} \text{Normal}(\theta, (\sqrt{\theta})^2)$, with variance θ . Find the UMVUE of $\theta(1 + \theta)$.

5. Let X_n be the outcome of the n -th toss of a coin with $P(\{H\}) = 0.75$, $n \in \mathbb{N}$. Can you find the approximate probability that after $N = 900$ tosses at least 450 heads are observed?

6. Let $X_1, \dots, X_n \stackrel{i.i.d.}{\sim} N(\mu, \sigma^2)$. Choose an appropriate pivot $T_n(\sigma^2)$ based on the MLE of σ^2 , and the point a such that $P(T_n(\sigma^2) < a) = 1 - \alpha$. Hence find a one-sided $1 - \alpha$ -confidence interval for σ^2 .

7. Consider the testing problem $H_0 : X \sim \text{exponential}(\lambda)$ against $H_1 : X \sim \text{Gamma}(k, \lambda)$ for some $k > 1$. Derive a most powerful level- α test for testing H_0 vs H_1 based on one sample.

Rough work

Rough work

Rough work